

1) Appendix

1.1) Matrix Algebra

Hermitian Matrix

A complex matrix A is said to be hermitian if $A^* = A$.

Note that $A^* = (\overline{A})^T$, that is if $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ then $A^* = \begin{pmatrix} \overline{a_{11}} & \overline{a_{21}} \\ \overline{a_{12}} & \overline{a_{22}} \end{pmatrix}$, $a_{ij} \in \mathbb{C}$.

Unitary Matrix

A complex matrix U is said to be unitary if $UU^* = U^*U = I$, moreover

- U is non singular and $U^* = U^{-1}$
- U has orthonormal rows and columns
- U is a rotation matrix in $y = Ux$ and thus $|y| = |x|$

Fourier Matrix

A fourier matrix is a $N \times N$ unitary matrix where the (i, j) th entry equals $e^{j2\pi ij/N}$, moreover it is scaled by a $1/\sqrt{N}$ factor. This matrix can be used to calculate the DFT or IDFT:

$$\text{DFT: } \mathbf{x} = \sqrt{N}U^*x \quad \text{IDFT: } x = \frac{1}{\sqrt{N}}U\mathbf{x}$$

Toeplitz and Circulant Matrix

A Toeplitz matrix has constant values along its diagonals, while a circulant matrix is described by only one row.

Singular Value Decomposition

Any $N \times M$ matrix can be factored as $A = U\Sigma V^*$ where

- U, V are unitary $N \times N, M \times M$ matrices
- Σ is a $N \times M$ matrix with real values on its main diagonal and 0 elsewhere

this is used for [precoding](#).

Matrix Inversion Lemma

Used for MMSE estimation of a complex Gaussian vector.

$$(A + BCD)^{-1} = A^{-1} - A^{-1}B(C^{-1} + DA^{-1}B)^{-1}DA^{-1}$$

this is used for [vector complex gaussian MMSE estimation](#)

1.2) Important Basic Formulas:

$$(AB)^{-1} = B^{-1}A^{-1}$$

$$\begin{aligned} \text{rank}(A + B) &\leq \text{rank}A + \text{rank}B \\ \|A\|_F &= \sqrt{\text{tr}(AA^*)} = \sqrt{\sum_i \sum_j |[A]_{i,j}|^2} \end{aligned}$$

Determinant:

$$\det(AB) = \det(BA) = \det A \det B$$

$$\det(I + AB) = \det(I + BA)$$

$$\det(A^{-1}) = 1/\det A$$

$$\det(e^A) = e^{\text{tr}A}$$

$$\det A = 0 \implies A^{-1} \text{ doesn't exist}$$

Trace:

$$\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA)$$

$$\text{tr}(aA + bB) = a\text{tr}A + b\text{tr}B$$

$$AB = a \implies \text{tr}(AB) = a$$

$$\mathbb{E}[ABC] = \text{tr}(\mathbb{E}[ABC]) = \mathbb{E}[\text{tr}(ABC)]$$

1.3) (Exam Candidate!) Moore Penrose Pseudo Inverse

Definition 1 (Moore Penrose Pseudo Inverse).

The pseudo inverse of a $N \times M$ matrix A is the unique matrix $A^\dagger$ satisfying:

$$AA^\dagger A = A$$

$$A^\dagger AA^\dagger = A^\dagger$$

such that $AA^\dagger, A^\dagger A$ are both hermitian. $((AA^\dagger)^* = AA^\dagger, (A^\dagger A)^* = A^\dagger A)$

- **Tall Matrix:** If A^*A is invertible then (left inverse)

$$A^\dagger = (A^*A)^{-1}A^*$$

this might be the case for $N \geq M$ ($A^\dagger A = I$)

- **Fat Matrix:** If AA^* is invertible then (right inverse)

$$A^\dagger = A^*(AA^*)^{-1}$$

this might be the case for $N \leq M$ ($AA^\dagger = I$)

- **Square Matrix:** If the matrix is square (and invertible) the pseudo inverse is the regular inverse.

Theorem 2 (Characterization of Least-Squares Solutions).

$\forall x \in \mathbb{K}^n$ we have $\|Ax - b\|_2 \geq \|Az - b\|_2$ where $z = A^\dagger b$ and $\|\cdot\|_2$ the euclidean norm.

The equality holds iff $x = A^\dagger b + (I - A^\dagger A)\vec{w}, \forall \vec{w}$ and provides infinite solutions unless A has full column rank (in this case $I - A^\dagger A = 0$). The solution with minimum euclidean norm is z .

For linear systems $Ax = b$ with non unique solutions, the pseudo inverse might be used to construct the solution of minimum euclidean norm among all solutions.

This is the **least squares problem**: $z = A^\dagger b = \min_x \|Ax - b\|_2$.

Theorem 3 (Minimum Norm Solution).

If $Ax = b$ is satisfiable, the vector $z = A^\dagger b$ is a solution and satisfies $\|z\|_2 \leq \|x\|_2$ for all solutions

$Ax = b$	Non-Satisfiable (Overdetermined)	Satisfiable (Underdetermined or 1 Solution)
Solutions	$Az - b = a$	$Az - b = 0$
Comment on z	z solves least squares	z is the solution with minimum norm

Proof:

- A^*A : $AA^\dagger A = A \underbrace{(A^*A)^{-1}}_{X^{-1}} \underbrace{A^*A}_X = AI = A$
- AA^* : $AA^\dagger A = \underbrace{AA^*}_X \underbrace{(AA^*)^{-1}}_{X^{-1}} A = IA = A$
- $A^\dagger = A^{-1}$: $AA^\dagger A = AA^{-1}A = AI = A$

used for [ZF equalization](#), [channel estimation](#)

1.4) Random Variables and Processes

Median PDF

The median of a pdf is simply $m : P(X < m) = \int_{-\infty}^m = 0.5$

- **Useful Formulas**

Bayes Formula:

$$\begin{aligned} f_{y|x}(y|x) &= \frac{f_{xy}(xy)}{f_x(x)} \\ f_{x|y}(y|x) &= \frac{f_{xy}(xy)}{f_y(y)} \end{aligned} \implies f_{x|y}(x|y) = \frac{f_{y|x}(y|x)f_x(x)}{f_y(y)}$$

Properness:

A complex random scalar x is said to be proper if

$$\begin{aligned} \mathbb{E}[x^2] &= \mathbb{E}[x]^2 \\ \mathbb{E}[xx^T] &= \mathbb{E}[x]\mathbb{E}[x^T] \end{aligned}$$

that is $\Im, \Re$ are uncorrelated and with same variance

Any subvector of a proper vector is proper:

$$x = [x_1, x_2]^T \text{ proper} \implies \begin{matrix} x_1 \text{ proper} \\ x_2 \text{ proper} \end{matrix}$$

however the opposite doesn't hold

$$\begin{matrix} x_1 \text{ proper} \\ x_2 \text{ proper} \end{matrix} \not\Rightarrow x = [x_1, x_2]^T \text{ proper}$$

Circular Symmetry

The distribution of $(x - \mathbb{E}[x])e^{j\phi}$ remains unchanged $\forall \phi$.

- **Linear Transformations**

$$y = ax \rightarrow f_y(y) = \frac{1}{|a|^2} f_x\left(\frac{y}{a}\right)$$

$$y = Ax \rightarrow f_y(y) = \frac{f_x(A^{-1}y)}{|\det A|^2}$$

Jensen Inequality

if $f(x)$ is convex, then

$$f(\mathbb{E}[x]) \leq \mathbb{E}[f(x)]$$

🕒 **Todo**

Convergences, Central Limit, LLN

2) Gaussian Distributions

Here are some relevant properties:

- $\perp \iff R = 0$
- If proper complex and gaussian then also circular, that is for complex gaussian Proper $\equiv$ Circular
- Gaussianity is preserved through linear transformations

Type	Real Scalar
Distribution	$f_x(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x-\mu)^2}{2\sigma^2} \right\}$
Differential Entropy	$h(x) = \frac{1}{2} \log_2 \pi e \sigma^2$

Type	Complex Scalar	Complex Vector
Distribution	$f_x(x) = \frac{1}{\pi\sigma^2} \exp \left\{ -\frac{\ x-\mu\ ^2}{\sigma^2} \right\}$	$f_x(x) = \frac{1}{\det(\pi R)} \exp \left\{ -(x-\mu)^* R^{-1} (x-\mu) \right\}$
Differential Entropy	$h(x) = \log_2 \pi e \sigma^2$	$h(x) = \log_2 \det(\pi e R)$
Mutual Information (ρ is SNR)	$I(\rho) = \log_2(1 + \rho)$ Small ρ: $I(\rho) = (\rho - \frac{1}{2}\rho^2) \log_2 e + o(\rho^2)$ Large ρ: $I(\rho) = \log_2 \rho + O(1/\rho)$	$I(\rho) = \log_2 \det(I + \rho A R_x A^* R_z^{-1})$
Sum $z = x + y$	$z \sim \mathcal{N}_{\mathbb{C}}(\mu_x + \mu_y, \sigma_x^2 + \sigma_y^2 + 2\Re \text{Cov}(x, y))$	if independent $z \sim \mathcal{N}_{\mathbb{C}}(\mu_x + \mu_y, R_x + R_y)$
AWGN	Complex Scalar	Complex Vector
Default form	$y = \sqrt{\rho} s + z$ $s \sim \mathcal{N}_{\mathbb{C}}(0, 1) \perp z \sim \mathcal{N}_{\mathbb{C}}(0, 1)$	$\mathbf{y} = \sqrt{\rho} \mathbf{A} s + \mathbf{z}$ $\mathbf{s} \sim \mathcal{N}_{\mathbb{C}}(0, R_x) \perp \mathbf{z} \sim \mathcal{N}_{\mathbb{C}}(0, R_z)$
Final y distribution:	$y \sim \mathcal{N}_{\mathbb{C}}(0, 1 + \rho)$	$y \sim \mathcal{N}_{\mathbb{C}}(0, \rho A R_s A^* + R_z)$
Final Likelihood $y s$ distribution	$y s \sim \mathcal{N}_{\mathbb{C}}(\sqrt{\rho} s, 1)$ $f_{y s}(y s) = \frac{1}{\pi} \exp \left\{ -\ y - \sqrt{\rho} s\ ^2 \right\}$	$y s \sim \mathcal{N}_{\mathbb{C}}(\sqrt{\rho} A s, R_z)$ $f_{y s}(y s) = \frac{1}{\pi^{N_y} \det(R_z)} \exp \left\{ -(y - \sqrt{\rho} A s)^* R_z^{-1} (y - \sqrt{\rho} A s) \right\}$

Type	Complex Scalar	Complex Vector
Final posteriori $s_{ y}$ distribution	$f_{s_{ y}} \propto f_{y_{ s}} \cdot f_s \sim \mathcal{N}_{\mathbb{C}}(\frac{\sqrt{\rho}}{1+\rho}y, \frac{1}{1+\rho})$	Mean: $\mu_{s_{ y}} = \sqrt{\rho}R_sA^*(\rho AR_sA^* + R_z)^{-1}\mathbf{y}$ Variance: $R_{s_{ y}} = (R_s^{-1} + \rho A^*R_z^{-1}A)^{-1}$

3) A Primer on Information Theory and MMSE

3.1) Information Content

Information

Information is defined as

$$i(A) = \log_2 \left(\frac{1}{P[A]} \right) [bit]$$

Given a discrete r.v. with alphabet $\mathcal{A}_x = \{a_1, \dots, a_M\}$ it is possible to describe a information function

$$i_x : \mathcal{A}_x \rightarrow \mathbb{R}, i_x(a) = i(x = a) = \log_2 \frac{1}{p_x(a)}$$

Entropy

$$\mathcal{H}(x) = - \sum_x p_x(x) \log_2 p_x(x) = -\mathbb{E}[\log_2 p_x(x)]$$

Properties:

- non negative
- PMF that maximizes H is uniform distribution
- Shift: $\mathcal{H}(x + c) = \mathcal{H}(x)$
- Additional Info reduces Entropy: $\mathcal{H}(x|y) \leq \mathcal{H}(x)$
- Let x have cardinality M , then $H(x) \leq \log_2 M$
- Let x_i be independent, then $H(\mathbf{x}) = \sum_i H(x_i)$

Joint and Conditional entropy

Joint:

$$\mathcal{H}(x_0, x_1) = \mathcal{H}(x_1) + \mathcal{H}(x_0|X_1)$$

$$\mathcal{H}(x_0, \dots, x_{N-1}) = \sum_{i=0}^{N-1} \mathcal{H}(x_i | x_0, \dots, x_{i-1})$$

Conditional:

$$\mathcal{H}(x|y) = - \sum_x \sum_y p_{xy}(xy) \log_2 p_{x|y}(x|y)$$

Differential Entropy

$$h(x) = - \int_x f_x(x) \log_2 f(x) dx = -\mathbb{E}[\log_2 f_x(x)]$$

Properties:

- Doesn't have physical meaning. IT CAN BE NEGATIVE
- Gaussian distribution maximizes it
- Used for mutual inf calculations
- Cannot be approached by discretizing $f(x)$

Relative Entropy

Consider two PMFs $p(\cdot)$, $q(\cdot)$. If $q(\cdot) \neq 0 \in \text{support of } p(\cdot)$, the their relative entropy is

$$D(p||q) = \sum_x p(x) \log_2 \frac{p(x)}{q(x)} = \mathbb{E} \left[\log_2 \frac{p(x)}{q(x)} \right]$$

Note that his is not symmetric $D(q||p) \neq D(p||q)$. This is interpreted as a similarity measure between q and p, and it is 0 iff $p = q \forall x$.

Moreover for PDFs:

$$D(f||g) = \int f(x) \log_2 \frac{f(x)}{g(x)} dx$$

Mutual Information

$$\begin{aligned} I(s; y) &= H(s) - H(s|y) \\ &= H(y) - H(y|x) \\ &= D(p_{sy} || p_s p_y) \end{aligned}$$

In contrast with differential entropies, $I(s, y)$ can be obtained by the limit of discretized version of s, y , albeit the differential entropies diverge, their difference is still well behaved.

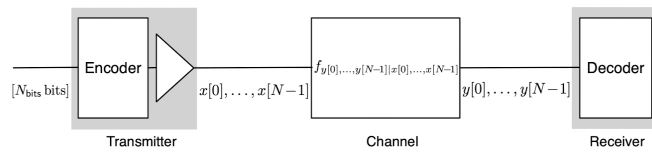
Moreover the mutual information satisfies the **chain rule**:

$$I(x_0, \dots, x_{N-1}; y) = \sum_{n=0}^{N-1} I(x_n; y | x_0, \dots, x_{N-1})$$

3.2) Reliable Communications

Channel Law

Let's start by explaining the communication link:



Basic Communication Link

- The encoder parses N_b bits into messages. These bits are IID and bernoulli.
- Each message is a collection of N_b bits and therefore there exist 2^{N_b} messages.
- Each message is mapped into N unit power complex symbols $s[0], \dots, s[N-1]$
- Each symbol is then amplified $x[0], \dots, x[N-1]$
- The channel randomly modifies the signal. The **channel is known** if the conditional probability $f_{y|x}$ ($f_{y[0], \dots, y[N-1] | x[0], \dots, x[N-1]}$) or $f_{y|x}(y|x)$ is known. Since the power amplification is known, this directly allows to find $f_{y|s}$.
- The decoder maps the observed values to the correct (ideally) codeword

Knowing the channel law, mutual information becomes:

$$I(\bar{s}, \bar{y}) \stackrel{\text{memoryless}}{=} \mathbb{E} \left[\log_2 \frac{f(\bar{y}|\bar{s})}{f(\bar{y})} \right] \stackrel{\text{stationary}}{=} \mathbb{E} \left[\log_2 \frac{f(\bar{y}|\bar{s})}{\frac{1}{N_{bits}} \sum_{i=0}^{N_{bits}-1} f(\bar{y}|\bar{s})} \right]$$

Capacity

Capacity is the **maximum rate** at which information can be transmitted over a channel with **arbitrarily small error probability**. That is, there exists a code such that, $p_e \rightarrow 0, N \rightarrow \infty$.

Therefore, since infinitely long signals aren't physically realizable we have that

$$\frac{R}{B} \leq C$$

R/B is the **spectral efficiency** measured in $\left[\frac{\text{bit/s}}{\text{Hz}} \right]$.

$$C_{\text{stationary, ergodic, memoryless}} = \max I(s; y)$$

$$C_{\text{ergodic, stationary}} = \max \lim_{N \rightarrow \infty} \frac{1}{N} I(\bar{s}; \bar{y})$$

This if the channel is information stable. And the symbols $s[n]$ are zero mean.

In an arbitrary channel we have the following probability to make an error:

$$p_e = \sum_{m=0}^{2^{N-b}-1} P[\hat{w} \neq m | w = m] P[w = m] \stackrel{\text{equiprobable}}{=} \frac{1}{2^{N_b}} \sum_{m=0}^{2^{N-b}-1} P[\hat{w} \neq m | w = m]$$

Information Stable

If a channel is information stable, then C is the maximum mutual information.

$$i(\bar{s}; \bar{y}) = \log_2 \frac{f(\bar{s}, \bar{y})}{f(\bar{s})f(\bar{y})}$$

A channel is information stable if information density does not deviate asymptotically from mutual information.

$$\lim_{n \rightarrow \infty} \frac{1}{N} i(\bar{s}, \bar{y}) = \lim_{n \rightarrow \infty} \frac{1}{N} I(\bar{s}, \bar{y})$$

and this applies to ergodic and stationary channels.

Coding Decoding

✓ Todo

(! IMPORTANT) Since capacity is only reached for infinite length codewords which are not achievable, there is a performance penalty on the capacity:

$$\frac{R}{B} = C - \underbrace{\sqrt{\frac{V}{N}} Q^{-1}(p_e)}_{\text{penalty!}} + O\left(\frac{\log N}{N}\right)$$

3.3) MMSE

Old Estimators

MAP: $\hat{s} = \arg \max_s f_{s|y}(s|y)$

ML: $\hat{s} = \arg \max_s f_{y|s}(y|s) \Rightarrow$ ML id IID

The MSE $|s - \hat{s}|^2$ is a measure of the estimators.

Min Mean Square Error (MMSE)

The MMSE Estimator is the estimation of s given y that minimizes the MSE:

$MMSE(s(\cdot)) = \arg \min_{\hat{s}(\cdot)} \mathbb{E}[|s - \hat{s}(\cdot)|^2]$. And the estimator has the following solution

$$\hat{s}(\cdot) = \mathbb{E}[s|\cdot]$$

Thus

$$MMSE(s(\cdot)) = \mathbb{E}[|s - \mathbb{E}[s|y]|^2] = \mathbb{E}[\text{var}(s|y)]$$

Theorem 4 (Properties of estimator).

- **Unbiased for the mean:**

$$\mathbb{E}[\hat{s}(y)] = \mathbb{E}[\mathbb{E}[s|y]] = \mathbb{E}[s]$$

- **Biased for specific realization**

$$\mathbb{E}[\hat{s}|s] = \mathbb{E}[(s|y)|s] \neq s$$

- **Orthogonality Principle:** error is orthogonal (uncorrelated) to any function $g(\cdot)$ of observation y

$$\mathbb{E}[g(y^*)(s - \hat{s})] = 0$$

- **Decreasing function:** given f_s a signal format and f_z a noise distribution, then MMSE is a decreasing function of SNR ρ

Proof:

$$\mathbb{E}[\mathbb{E}[s|y]] = \int \left(\int s \cdot p(s|y) ds \right) p(y) dy = \int s \left(\int p(s|y) p(y) dy \right) ds = \int s \cdot p(s) ds = \mathbb{E}[s]$$

$$\begin{aligned} E[g(y)(s - \hat{s})] &= \mathbb{E}[\mathbb{E}[g(y)(s - \hat{s})|y]] = \mathbb{E}[\mathbb{E}[g(y)s - g(y)\hat{s}|y]] \\ &= \mathbb{E}[\mathbb{E}[g(y)s|y] - \mathbb{E}[g(y)\hat{s}|y]] = \mathbb{E}[g(y)(\mathbb{E}[s|y] - \mathbb{E}[\hat{s}|y])] \\ &= \mathbb{E}[g(y)(\mathbb{E}[s|y] - \mathbb{E}[s|y])] = \mathbb{E}[g(y) \cdot 0] = \mathbb{E}[0] \\ &= 0 \end{aligned}$$

$$\text{Since: } \mathbb{E}[\hat{s}|y] = \mathbb{E}[\mathbb{E}[s|y]|y] = \mathbb{E}[s|y]$$

$$\text{In general recall law of total expectation: } \mathbb{E}[f(y)] = \mathbb{E}[\mathbb{E}[f(y)|y]], \mathbb{E}[f(y)|y] = \mathbb{E}[\mathbb{E}[f(y)|y]|y]$$

MMSE With Known Channel

If the channel is known we have $f_{y|s}$ known, then

$$f_{s|y}(s|y) = \frac{f_{y|s}(y|s)f_s(s)}{f_y(y)} = \frac{f_{y|s}(y|s)f_s(s)}{\int f_{y|s}(y|s)f_s(s)ds}$$

and therefore the estimator is

$$\begin{aligned} \hat{s}(y) &= \mathbb{E}[s|y] = \int s f_{s|y}(s|y) ds = \int \frac{s f_{y|s}(y|s)f_s(s)}{\int f_{y|s}(y|s)f_s(s)ds} ds \\ &= \frac{\int s f_{y|s}(y|s)f_s(s)ds}{\int f_{y|s}(y|s)f_s(s)ds} \end{aligned}$$

MMSE in AWGN

Complex scalar:

Recall $y = \sqrt{\rho}s + z$ with $s, z \sim \mathcal{N}_{\mathbb{C}}(0, 1)$, then $f_s = \frac{1}{\pi} \exp\{-|s|^2\}$ and $f_{y|s} = \frac{1}{\pi} \exp\{-|y - \sqrt{\rho}s|^2\}$.

$$\hat{s}(y) = \frac{\sqrt{\rho}}{1 + \rho} y$$

where it was found that

$$f_{s|y} \propto f_{y|s} \cdot f_s \sim \mathcal{N}_{\mathbb{C}}\left(\frac{\sqrt{\rho}}{1 + \rho}, \frac{1}{1 + \rho}\right)$$

and the MMSE is

$$MMSE(\rho) = \frac{1}{1 + \rho}$$

In fact, $\rho \rightarrow \infty \implies MMSE \rightarrow 0$.

And has a bias of

$$-\frac{1}{1 + \rho} s$$

Vector MMSE

$$\hat{s}(y) = \mathbb{E}[s|y]$$

$$E = \mathbb{E}[(s - \hat{s}(y))(s - \hat{s}(y))^*]$$

Where the j-th diagonal in E represents the MMSE for the j.th entry of s

Vector Gaussian MMSE

Very Easily:

$$\hat{s}(y) = \sqrt{\rho} R_s A^* (I + \rho A R_s A^*)^{-1} y$$

$$E = R_s - \rho R_s A^* (I + \rho A R_s A^*)^{-1} A R_s$$

By applying the [matrix inversion lemma](#) we obtain:

$$E = (R_s^{-1} + \rho A^* A)^{-1}$$

I-MMSE in gaussian noise

Scalar:

$$MMSE(\rho) = \frac{1}{\log_2 e} \frac{d}{d\rho} I(s; \sqrt{\rho}s + z) \quad \frac{1}{\log_2 e} I(\rho) = \int_0^\rho MMSE(\xi) d\xi$$

Vector:

$$\frac{1}{\log_2 e} \nabla_A I(s; \sqrt{\rho} A s + z) = \rho A E$$

L-MMSE

works for gaussian vectors in gaussian noise

$$\hat{s} = W^{MMSE*} y + b^{MMSE}$$

where b is the bias term for non zero mean signals. It is therefore usually ignored.

$$W^{MMSE} = R_y^{-1} R_{ys}$$

from here the MMSE estimate is:

$$E = \mathbb{E}[(s - \hat{s})(s - \hat{s})^*] = R_s - R_{\hat{s}s} - R_{\hat{s}}^* + R_{\hat{s}} = R_s - R_{ys}^* R_y^{-1} R_{ys}$$

in AWGN we have

$$W^{MMSE} = \sqrt{\rho} \underbrace{(R_z + \rho A R_s A^*)}_{R_y^{-1}} \cdot \underbrace{A R_s}_{R_{ys}}$$

$$E = (R_s^{-1} + \rho A^* R_z^{-1} A)^{-1}$$

if both are scalars we turn to the MSE definition

Random Processes

TODO

3.4) Exercises (Textbook & Professor)

Example 1.2 (Entropy Of Bernoulli RV).

Express entropy of bernoulli rv:

$$x = \begin{cases} 0 & \text{with probability } p \\ 1 & \text{with probability } 1 - p \end{cases}$$

Solution:

Recall entropy: $H(x) = -\sum_x p(x) \log_2 p(x)$, then

$$H(x) = -p \log_2 p - (1 - p) \log_2 (1 - p)$$

Solve additional min/max problem:

$$\frac{d}{dp} H(x) = -\log_2(p) - \frac{1}{\ln p} + \log_2(1 - p) + \frac{1}{\ln(1-p)} = \log_2 \frac{1-p}{p}$$

where $p = 0.5$ provides the max entropy $H_{\max} = 1$ and the min entropy is $p = 0 \implies H_{\min} = 0$ by changing the log entropy formula

Some derivatives:

$$\begin{aligned} \frac{d}{dx} \log_2 x &= \frac{1}{x} \frac{1}{\ln 2} \\ \frac{d}{dx} \log_k f(x) &= \frac{f'(x)}{f(x)} \frac{1}{\ln k} \\ \frac{d}{dx} f(x) \log_k f(x) &= f'(x) \left(\log_k f(x) + \frac{1}{\ln k} \right) \end{aligned}$$

Example 1.4 (Negative Differential Entropy).

A uniform distribution is a textbook example of a possible negative differential entropy:

Let $x \sim u(0, b)$, then:

$$h(x) = -\int_0^b \frac{1}{b} \log_2 \frac{1}{b} dx = \log_2 b < 0, \forall b \in (0, 1)$$

Example 1.5 (Differential Entropy of complex Gaussian Scalar).

Let $x \sim \mathcal{N}_{\mathbb{C}}(\mu, \sigma^2)$

$$h(x) = -\mathbb{E}[\log_2(\frac{1}{\pi\sigma^2} e^{-|x-\mu|^2/\sigma^2})] = \mathbb{E}[\log_2 \pi\sigma^2] + \mathbb{E}\left[\frac{|x-\mu|^2}{\sigma^2}\right] \log_2 e = \log_2(\pi e \sigma^2)$$

Example 1.6 (Differential Entropy of complex Gaussian Vector).

Let $x \sim \mathcal{N}_{\mathbb{C}}(\mu, \mathbf{R})$, then

Recall $f_x(x) = \frac{1}{\det(\pi \mathbf{R}_x)} e^{-(x-\mu_x)^* \mathbf{R}_x^{-1} (x-\mu_x)} = \frac{1}{\det(\pi \mathbf{R})} e^{-k}$

Now from the definition of the differential entropy:

$$\begin{aligned} h(x) &= -\mathbb{E}[\log_2 f_x(x)] = -\mathbb{E}\left[\log_2\left(\frac{1}{\det(\pi \mathbf{R})} e^{-k}\right)\right] = -\mathbb{E}[-\log_2 \det(\pi \mathbf{R}) - k \cdot \log_2 e] \\ &= \log_2(\det \pi \mathbf{R}) + \mathbb{E}[k] \log_2 e = \log_2(\det \pi \mathbf{R}) + \mathbb{E}[(x - \mu_x)^* \mathbf{R}_x^{-1} (x - \mu_x)] \log_2 e \end{aligned}$$

Analyze just $\mathbb{E}[k]$:

$$\begin{aligned}
\mathbb{E}[k] &= \mathbb{E}[\text{tr}(k)] = \mathbb{E}[\text{tr}((x - \mu)^* R^{-1} (x - \mu))] \\
&= \mathbb{E}[\text{tr}(R^{-1} (x - \mu)(x - \mu)^*)] = \text{tr}(\mathbb{E}[R^{-1} (x - \mu)(x - \mu)^*]) \\
&= \text{tr}(\mathbb{E}[R^{-1} R]) = \text{tr}(I)
\end{aligned}$$

Now since $\det cI_n = c^n$ we have:

$$E[k] \log_2 e = \text{tr}(I) \log_2 e = n \log_2 e = \log_2 e^n = \log_2 \det eI$$

Finally:

$$h(x) = \log_2(\det \pi R) + \log_2 \det eI = \log_2 \det(\pi R e I) = \log_2 \det(\pi e R)$$

Example 1.5 (Gaussian Mutual Information for complex gaussian scalar).

Suppose a standard AWGN channel $y = \sqrt{\rho}s + z$, and:

- $s \sim \mathcal{N}_{\mathbb{C}}(0, 1)$
- $z \sim \mathcal{N}_{\mathbb{C}}(0, 1)$
- then
- $y \sim \mathcal{N}_{\mathbb{C}}(0, 1 + \rho)$
- $y|s \sim \mathcal{N}_{\mathbb{C}}(\sqrt{\rho}s, 1)$
- therefore

$$\begin{aligned}
I(\rho) &= I(s; \sqrt{\rho}s + z) = h(\sqrt{\rho}s + z) - h(\sqrt{\rho}s + z|s) \\
&= h(y) - h(z) = \log_2(\pi e(1 + \sqrt{\rho})) - \log_2(\pi e) = \log_2 1 + \rho
\end{aligned}$$

Example 1.6 (Gaussian Mutual Information for complex gaussian vector).

Exactly identical as 1.6, check the table on gaussian distributions for infos

Example 1.16 (Channel Law in AWGN Channel).

Let $y[n] = \sqrt{\rho}s[n] + z[n]$ with $z \sim \mathcal{N}_{\mathbb{C}}(0, 1)$

Recall that $y|s \sim \mathcal{N}_{\mathbb{C}}(\sqrt{\rho}s, 1)$ and thus

$$f_{y|s}(y|s) = \frac{1}{\pi} \exp \{ -|y - \sqrt{\rho}s|^2 \}$$

🔄 **Todo**

Example 1.35

🔄 **Todo**

Derivation of LMMSE

Exercise (capacity of BSC).

The binary channel in which the input symbols are complemented with probability p is the simplest model of a channel with errors, yet it captures most of the complexity of the general problem. Calculate the capacity

Solution:

let $p_s(s) = \begin{cases} 0 & \text{with probability } p_s \\ 1 & \text{with probability } 1 - p_s \end{cases}$, then the channel law is: $p_{y|s}(y|s)$ and can be found in this way:

$$p_{y|s}(y|s) \begin{cases} p(0|s) = \begin{cases} p(0|0) = 1 - p \\ p(1|0) = p \end{cases} \\ p(1|s) = \begin{cases} p(1|0) = p \\ p(1|1) = 1 - p \end{cases} \end{cases}$$

and then the capacity is the maximum of the mutual information $I(s; y) = H(y) - H(y|s)$ where $H(y) = 1$ is maximized when $p_s = 0.5$. Now we must only calculate $H(y|s) = p_s(0)H(y|0) + p_s(1)H(y|1)$ by symmetry the two entropies are the same, so I just calculate one of them:

$$-H(y|0) = p(0|0)\log_2 p(0|0) + p(1|0)\log_2 p(1|0) = (1 - p)\log_2(1 - p) + p\log_2(p)$$

$$H(s; y) = 1 - 0.5[2(H(y|s))] = 1 + p\log_2 p + (1 - p)\log_2(1 - p)$$

And thus

$$C = \max I(s; y) \rightarrow 1$$

Exercise 1.34 (Orthogonality Proof + MMSE Proof).

Consider the AWGN transformation: $y = \sqrt{N_0} + \rho s + z$:

- Prove the orthogonality principle: $\mathbb{E}[g(y)(s - \mathbb{E}[s|y])] = 0$
- Prove that the MMSE is $\hat{s}(y) = \mathbb{E}[s|y]$

For the first solution watch the proof provided in the above chapter

The trick for the second part is to rewrite the square as:

$$\begin{aligned} |s - \hat{s}|^2 &= |(s - \mathbb{E}[s|y]) + (\mathbb{E}[s|y] - \hat{s})|^2 \\ &= |s - \mathbb{E}|^2 + |\mathbb{E} - \hat{s}|^2 + 2\Re[(s - \mathbb{E})(\mathbb{E} - \hat{s}^*)] \\ &= |s - \mathbb{E}|^2 + |\mathbb{E} - \hat{s}|^2 + 2\Re[g(y)(s - \mathbb{E})] \end{aligned}$$

where $g(y) = (\mathbb{E}[s|y] - \hat{s}(y))$. Now apply the expectation and the last term is the orthogonality principle:

$$\mathbb{E}[|s - \hat{s}|^2] = \mathbb{E}[|s - \mathbb{E}|^2] + \mathbb{E}[|\mathbb{E} - \hat{s}|^2] + 0$$

and is minimized when $\hat{s}(y) = \mathbb{E}[s|y]$.

4) A Signal Processing Perspective

4.1) Signal, Channel and Noise Representations

Ideal Passband Signal

$$\begin{aligned}
 x_p(t) &= A(t) \cos(2\pi f_c t + \phi(t)) \\
 &= \underbrace{A(t) \cos(\phi(t))}_{\sqrt{2}x_i(t)} \cos(2\pi f_c t) - \underbrace{A(t) \sin(\phi(t))}_{\sqrt{2}x_q(t)} \sin(2\pi f_c t) \\
 &= \sqrt{2}x_i(t) \cos(2\pi f_c t) - \sqrt{2}x_q(t) \sin(2\pi f_c t)
 \end{aligned}$$

where $\sqrt{2}$ is used for power conservation between passband and baseband:

$$x_p(t) = \sqrt{2} \Re \left\{ x(t) e^{j2\pi f_c t} \right\}$$

in frequency:

$$x_p(f) = \frac{1}{\sqrt{2}} [(x(f - f_c)x^*(f - f_c)]$$

Complex Baseband Equivalent

$$x(t) = x_i(t) + jx_q(t)$$

proof of the power conservation:

$$\begin{aligned}
 x_p(t) &= \sqrt{2} \Re \{ (x_i + jx_q)(\cos(2\pi f_c t) + j \sin(2\pi f_c t)) \} \\
 &= \sqrt{2} \Re \{ x_i \cos - x_q \sin + j(x_i \sin + x_q \cos) \} \\
 &= \sqrt{2} [x_i(t) \cos(2\pi f_c t) - x_q(t) \sin(2\pi f_c t)]
 \end{aligned}$$

Downconversion

$$y_p = \sqrt{2} (y_i(t) \cos(2\pi f_c t) - y_q(t) \sin(2\pi f_c t))$$

and the complex baseband equivalent is found by multiplying and then setting a low pass filter:

- $\sqrt{2} \cos(2\pi f_c t) \rightarrow y_p(\sqrt{2} \cos(2\pi f_c t)) = \underline{y_i(t)} + y_i(t) \cos(4\pi f_c t) - y_q(t) \sin(2\pi f_c t)$
 - $-\sqrt{2} \sin(2\pi f_c t)$ analogously finds $y_q(t)$
- recombine into $y(t) = y_i(t) + jy_q(t)$

Complex Baseband Channel

Let $c_p(t)$ be the complex passband channel, then the input and output relation becomes:

$$y_p(t) = (c_p * x_p)(t) = \int_{-\infty}^{\infty} c_p(\tau) x_p(t - \tau) d\tau$$

Note that x_p doesn't need to be band limited since x_p already is. For the same reason, also the output is band limited.

The baseband channel is given by

$$g_{B/2}(\tau) = B \text{sinc}(B\tau) \rightarrow \boxed{c_b(\tau) = g_{B/2}(\tau) * [c_p(\tau) e^{-j2\pi f_c \tau}]}$$

Complex Pseudo-Baseband Channel

This is the downshifted but not band limited version of the baseband channel. This equals to just

$$\boxed{c(\tau) = c_p(\tau) e^{-j2\pi f_c \tau}}$$

This is useful for single tap discrete channels.

Since this is not band limited it cannot be sampled and care must be exercised when time discretizing.

Tapped Delay Line

$$c_p(\tau) = \sum_i A_i \delta(\tau - \tau_i)$$

This is a sum of deltas and therefore it is not band limited

$$c_b(\tau) = g_{B/2} * \underbrace{\sum A_i \delta(\tau - \tau_i) e^{-j2\pi f_c \tau_i}}_{c(\tau)}$$

Time Discretization

The critical sampling period is the equality:

$$T \leq \frac{1}{B}$$

and therefore

$$c[l] = T c_b(\tau)|_{\tau=lT+\phi} = T c_b(lT + \phi)$$

where $\phi = 0$ usually but can be set to any constant to **offset** the sampling. This is also non band limited.

It is not necessary to find c_b to obtain $c[l]$ since it is already obtainable from the pseudo baseband downshifted not band limited) $c(\tau)$.

- **Pulse Shaping**

a continuous time signal can be represented as (in general $g_{tx}(t - nT)$, not sinc)

$$x(t) = \sum_n x[n] \text{sinc}\left(\frac{t - nT}{T}\right)$$

then $g_{B/2}$ is the convolution of the two tx and rx pulse shapes.

Raised Cosine Pulse: band penalty of $1/1 + b$.

AWGN Noise

- White since autocovariance $R_{v_p} = \mathbb{E}[v_p(t)v_p^*(t + \tau)] = N_0\delta(t)$ and therefore it is stationary and ergodic and has power spectral density $S_{v_p}(f) = N_0$
- Gaussian since $v_p \sim \mathcal{N}_{\mathbb{C}}(0, N_0)$ at each time t , then by the central limit theorem a sum of iid is also gaussian. Moreover $N_0 = kT_{eff}$.

Since bandlimited the noise has power $P = \int_B N_0 df = N_0 B$

Energy And Power

TODO

The avg transmit power is

$$P_t = \frac{1}{N} \lim_{N \rightarrow \infty} \sum_{n=0}^{N-1} \mathbb{E}[|x[n]|^2] [W]$$

But the limit can be ignored.

The average transmit energy per symbol is $E_s = P_t T [J]$. In one period the noise has energy N_0 and if both sequences x, v are scaled by $\sqrt{T}$ we have that

$$E_s = \frac{1}{N} \sum_{n=0}^{N-1} \mathbb{E}[|x[n]|^2]$$

By making $s[n]$ a unit variance (power) symbol, we have that

$$x[n] = \sqrt{E_s P[n]} s[n]$$

with $\frac{1}{N} \sum P[n] = 1$ to register variations.

Channel Normalization

$$\begin{aligned} c(\tau) &= \sqrt{G} h(\tau) \\ c_b(\tau) &= \sqrt{G} h_b(\tau) \\ c[l] &= \sqrt{G} h[l] \end{aligned}$$

where $\sum h[l] = 1$

$$\boxed{SNR = \frac{P_r}{N_0/T} = \frac{GE_s}{N_0}}$$

and so the AWGN relation becomes:

$$y[n] = \sqrt{G} \sum_{l=0}^L h[l] x[n-l] + v[n]$$

where the sum is a FIR filter, otherwise it would sum $-\infty, \infty$. If the channel is single tap (frequency selective), then it multiplicative.

Vector Representation

This convolution can be expressed as a matrix multiplication.

we can define the impulse response of the channel to be a **toeplitz matrix** of dimensions $N \times (N + L)$.

$$\overline{\mathbf{H}}_{N,N+L} = \begin{bmatrix} h[L] & \dots & h[0] & 0 & 0 & \dots & 0 \\ 0 & h[L] & \dots & h[0] & 0 & \dots & 0 \\ \vdots & & & \ddots & & & \vdots \\ 0 & \dots & \dots & 0 & h[L] & \dots & h[0] \end{bmatrix}$$

And also create a vector of symbols with memory of L past symbols.

$$\overline{\mathbf{x}}_{N+L} = \begin{bmatrix} x[-L] \\ \vdots \\ x[-1] \\ x[0] \\ \vdots \\ x[N-1] \end{bmatrix}$$

Then

$$\overline{\mathbf{y}} = \sqrt{G} \overline{\mathbf{H}}_{N,N+L} \overline{\mathbf{x}}_{N+L} + \overline{\mathbf{v}}_N$$

this represents all the N symbols received.

$$\overline{\mathbf{y}}_N = [y[0], \dots, y[N-1]]^T \quad \overline{\mathbf{v}}_N = [v[0], \dots, v[N-1]]^T$$

4.2) MIMO

This is just the previous model extended to N_t tx antennas (j) and N_r rx antennas (i).

$$y^{(i)}(t) = \sqrt{G} \sum_{j=0}^{N_t-1} \int_{\mathbb{R}} h^{(i,j)}(\tau) x^{(j)}(t - \tau) d\tau + v_b^{(i)}(t)$$

vectorized

$$\mathbf{y}(t) = \sqrt{G} \int_{-\infty}^{\infty} \mathbf{H}(\tau) \mathbf{x}(t - \tau) d\tau + \mathbf{v}(t)$$

where:

- $\mathbf{y}$ is the vector of all i rx signals $\mathbf{y}(t) = [y^{(0)}(t), \dots, y^{(N_r-1)}(t)]^T$
- $\mathbf{x}$ is the vector of all j tx signals $\mathbf{x}(t) = [x^{(0)}(t), \dots, x^{(N_t-1)}(t)]^T$
- $\mathbf{v}$ the noise on all i rx antennas $\mathbf{v}(t) = [v^{(0)}(t), \dots, v^{(N_r-1)}(t)]^T$
- $\mathbf{H}$ the matrix of all normalized channel responses.

$$\mathbf{H}(\tau) = \begin{bmatrix} h^{(0,0)}(\tau) & h^{(0,1)}(\tau) & \dots & h^{(0,N_t-1)}(\tau) \\ h^{(1,0)}(\tau) & h^{(1,1)}(\tau) & \dots & h^{(1,N_t-1)}(\tau) \\ \vdots & \vdots & \ddots & \vdots \\ h^{(N_t-1,0)}(\tau) & h^{(N_t-1,1)}(\tau) & \dots & h^{(N_t-1,N_t-1)}(\tau) \end{bmatrix}$$

and time discretized:

$$\mathbf{y}[n] = \sqrt{G} \sum_{l=0}^L \mathbf{H}[l] \mathbf{x}[n - l] + \mathbf{v}[n]$$

where

Stacked Vector Representation

again, the the full block of N tx signals received by N_r rx antennas can be written as a matrix multiplication by removing the convolution.

First for the single receiver:

$$\bar{\mathbf{y}}_N^{(i)} = \sqrt{G} \sum_{j=0}^{N_t-1} \bar{\mathbf{H}}_{N,N+L}^{(i,j)} \bar{\mathbf{x}}_{N+L}^{(j)} + \bar{\mathbf{v}}_N^{(i)}$$

where $\bar{\mathbf{y}}_n^{(i)} = [y_n^{(i)}(0), \dots, y_n^{(i)}(N-1)]$ $\bar{\mathbf{v}}_N^{(i)} = [v_n^{(i)}(0), \dots, v_n^{(i)}(N-1)]$
and most importantly:

$$\bar{\mathbf{x}}^{(j)} = [x^{(j)}[-L], \dots, x^{(j)}[-1], x^{(j)}[0], x^{(j)}[1], \dots, x^{(j)}[N-1]]^T$$

$$H^{(i,j)} = \begin{bmatrix} h^{(i,j)}[L] & \dots & h^{(i,j)}[0] & 0 & 0 & \dots & 0 \\ 0 & h^{(i,j)}[L] & \dots & h^{(i,j)}[0] & 0 & \dots & 0 \\ \vdots & & & \ddots & & & \vdots \\ 0 & \dots & \dots & 0 & h^{(i,j)}[L] & \dots & h^{(i,j)}[0] \end{bmatrix}$$

From here the stacking of these matrices can be used to also give the full received signals vector.

In particular

- $\mathbf{x}$: matrix of $(N + L)$ tx matrices (made of N_t antennas) $\rightarrow$ tot size $N_t(N + L)$
- $\mathbf{y}$: matrix of N rx matrices (made of N_r antennas) $\rightarrow$ tot size $N_r N$
- $\mathbf{v}$: same as $\mathbf{y}$ but for noise
- $\mathbf{H}$: matrix made of all toeplitz matrices of all i -j channels $\rightarrow$ tot size $N_r N \times N_t(N + L)$

$$\bar{\mathbf{x}}_{(N+L)N_t} = \begin{bmatrix} \bar{\mathbf{x}}_{N+L}^{(0)} \\ \vdots \\ \bar{\mathbf{x}}_{N+L}^{(N_t-1)} \end{bmatrix}, \quad \bar{\mathbf{H}}_{N_r N_r, (N+L)N_t} = \begin{bmatrix} \bar{\mathbf{H}}_{N.N+L}^{(0,0)} & \cdots & \bar{\mathbf{H}}_{N.N+L}^{(N_r,0)} \\ \vdots & \ddots & \vdots \\ \bar{\mathbf{H}}_{N.N+L}^{(0,N_t-1)} & \cdots & \bar{\mathbf{H}}_{N.N+L}^{(N_r,N_t-1)} \end{bmatrix}$$

finally:

$$\bar{\mathbf{y}}_{N_r N} = \sqrt{G} \bar{\mathbf{H}}_{N_r N, (N+L)N_t} \bar{\mathbf{x}}_{N_t+L} + \bar{\mathbf{v}}_{N_r N}$$

Form	Inputs	Outputs	What Happens	Formula
SISO conv	1	1	One signal through L taps FIR channel.	$y[n] = \sqrt{G} \sum_{l=0}^L h[l]x[n-l] + v[n]$
SISO matrix	$N + L$	N	Entire block of N signals through L taps FIR channel.	$\bar{\mathbf{y}} = \sqrt{G} \bar{\mathbf{H}}_{N.N+L} \bar{\mathbf{x}}_{N+L} + \bar{\mathbf{v}}_N$
MIMO scalar	N_t	1	I-th RX antenna receiving 1 signal from all TX antennas in continuous time channel.	$y^{(i)}(t) = \sqrt{G} \sum_{j=0}^{N_t-1} \int_{\mathbb{R}} h^{(i,j)}(\tau) x^{(j)}(t-\tau) d\tau + v_b^{(i)}(t)$
MIMO conv	N_t	N_r	All rx antennas from all tx antennas one signal. If frequency flat equivalent to frequency selective siso matrix.	$\mathbf{y}[n] = \sqrt{G} \sum_{l=0}^L \mathbf{H}[l] \mathbf{x}[n-l] + \mathbf{v}[n]$
MIMO stacked	$(N + L)N_t$	$N_r N$	Entire N signal transmit block across all antennas. Matrix size same as SISO matrix multiplied by N_t, N_r .	$\bar{\mathbf{y}}_{N_r N} = \sqrt{G} \bar{\mathbf{H}}_{N_r N, (N+L)N_t} \bar{\mathbf{x}}_{N_t+L} + \bar{\mathbf{v}}_{N_r N}$

MIMO Channel Normalization

Same as SISO case:

The MIMO channel are normalized jointly over all entries, that is

$$\sum_{l=0}^L \sum_{i=0}^{N_r-1} \sum_{j=0}^{N_t-1} \mathbb{E}[|H[l]_{i,j}|^2] = N_r N_t$$

Where the two summations over i and j can be compactly formulated with the **Frobenius** norm:

$$\sum_{l=0}^L \mathbb{E}[\|H[l]\|_F^2] = \sum_{l=0}^L \mathbb{E}[\text{tr}(HH^*)] = N_r N_t$$

Precoding

Recall the power decomposition of the SISO case

$$x[n] = \sqrt{E_s P[n]} \cdot s[n] \xrightarrow{\text{MIMO}} \mathbf{x}[n] = \sqrt{\frac{E_s}{N_t}} \mathbf{F}[n] \mathbf{s}[n]$$

with $N_s \leq N_t$, N_s the size of the $\mathbf{s}$ vector and thus the concurrent data streams. (if more signals than antennas information is lost, duh)

The codewords are IID:

$$\mathbb{E}[\mathbf{s}[n]\mathbf{s}^*[n]] = I$$

and $F[n]$ must be normalized

$$\frac{1}{N} \sum_{n=0}^{N-1} \|\mathbf{F}[n]\|_F^2 = \frac{1}{N} \sum_{n=0}^{N-1} \text{tr}(\mathbf{F}[n]\mathbf{F}^*[n]) = N_t$$

In this way the covariance is:

$$R_x = \mathbb{E}[xx^*] = \frac{E_s}{N_t} \mathbb{E}[Fss^*F^*] = \frac{E_s}{N_t} \mathbb{E}[FIF^*] = \frac{E_s}{N_t} F[n]F^*[n]$$

where the i-th diagonal holds the avg power of the i-th antenna and the off diagonals show the correlation between antenna $R_{x_{i,j}}$ i and j.

Case	SISO	MIMO
$s[n]$ power	unit power	All symbols: N_s Single symbol: unit power
$x[n]$ instantaneous power	$E_s P[n]$	Diagonal of $\frac{E_s}{N_t} F[n]F^*[n]$
average $x[n]$ power	E_s	Over all antennas: E_s

the [singular value decomposition](#) can be used to rewrite F :

$$F[n] = \underset{\substack{\uparrow \\ \text{Beam} \\ \text{Steering} \\ \text{Matrix}}}{U_F[n]} \underset{\substack{\uparrow \\ \text{Power} \\ \text{Allocation} \\ \text{Matrix}}}{\Sigma_F[n]} \underset{\substack{\uparrow \\ \text{Mixing} \\ \text{Matrix}}}{V_F^*[n]}$$

With:

- U_F, V_F unitary and respectively $N_t \times N_t, N_s \times N_s$
- $\Sigma_F = \begin{bmatrix} P^{1/2} \\ 0 \end{bmatrix} N_t \times N_s$ matrix where $P^{1/2}$ is a square diagonal $N_s \times N_s$ matrix with the values $P_0, \dots, P_{N_s-1}$

Signal constraints

Power constraints:

	SISO	MIMO
Per block	$\frac{1}{N} \sum_{n=0}^{N-1} \mathbb{E}[\ x[n]\ ^2] = E_s$	$\frac{1}{N} \sum_{n=0}^{N-1} \mathbb{E}[\ x[n]\ ^2] = E_s \iff \frac{1}{N} \sum_{n=0}^{N-1} \text{tr}(R_x) = E_s$
Per block in precoder	$\frac{1}{N} \sum_{n=0}^{N-1} P[n] = 1$	$\frac{1}{N} \sum_{n=0}^{N-1} \text{tr}(F[n]F^*[n]) = N_t \iff \frac{1}{N} \sum_{n=0}^{N-1} \text{tr}(P[n]) = N_t$
Per symbol	$\mathbb{E}[\ x[n]\ ^2] = E_s$	$\mathbb{E}[\ x[n]\ ^2] = E_s \iff \text{tr}(R_x) = E_s$
Per symbol in precoder	$P[n] = 1$	$\text{tr}(F[n]F^*[n]) = N_t \iff \text{tr}(P[n]) = N_t$
Per antenna	$[R_x] = E_s$	$[R_x]_{j,j} = \frac{E_s}{N_t}$

Magnitude:
TODO (?)

4.3) Linear Channel Equalization

Basic Formulation

Start with a scalar channel

$$w^*(z)\sqrt{G}h(z) = z^{-\Delta}$$

where Δ is the delay (in taps) as a design parameter.

The ideal equalizer is:

$$w^*(z) = \frac{1}{\sqrt{G}} \frac{1}{h(z)}$$

so that $(h * w^*)(z) = \delta(n - \Delta)$

This is applied to a FIR channel of length L and a FIR equalizer (L_{eq}) must be found.

MIMO Case

In the MIMO case the **left inverse of H that satisfies**

$$W^*(z)\sqrt{G}H(z) = D(z)$$

where is a diagonal matrix with the allowed delays $D(z) = \text{diag}(e^{-\Delta_0}, \dots, e^{-\Delta_{N_t-1}})$.

Special care must be put on $H(z)$.

Let $H(z) = \sum_{l=0}^L H[l]z^{-l}$ be the matrix containing the z transform of the i,j-th entry ($H[z]_{i,j} = h^{i,j}(0), \dots, h^{i,j}[L]$).

Theorem 5 (Theorem on FIR left inverses).

A MIMO transfer function $H(z)$ has a FIR left inverse iff $H(z)$ has full column rank $\forall z \neq 0$

It also turns out that:

(!exam candidate)

It turns out that an exact FIR ZF MIMO inverse exists for $N_r > N_t$ As long as the entries of H do not share a common root (coprime), then a FIR inverse exists

In the time domain the ZF eq is

$$\sqrt{G} \sum_{l=0}^{L_{eq}} W^*[l]H[n-l] = \text{diag}(\delta[n-\Delta_0], \dots, \delta[n-\Delta_{N_t-1}])$$

only the j-th row of W contributes to the eq of the j-th antenna. This implies that the analog expression for each row must hold on its own.

$$\sqrt{G} \sum_{l=0}^{L_{eq}} w_j^*[l]H[n-l] = d_j^*[n]$$

Written in the stacked vector representation we have that:

$$\begin{bmatrix} w_j[0] \\ \vdots \\ w_j[L_{eq}] \end{bmatrix} \xrightarrow{\sqrt{G} \bar{w}_j^* \bar{T}} \bar{d}_j^* \leftarrow \begin{bmatrix} d_j[0] \\ \vdots \\ d_j[L_{eq} + L] \end{bmatrix}$$

$$\begin{bmatrix} H[0] & \dots & H[L] & 0 & 0 & \dots & 0 \\ 0 & H[0] & \dots & H[L] & 0 & \dots & 0 \\ \vdots & & & \ddots & & & \vdots \\ 0 & \dots & 0 & 0 & H[0] & \dots & H[L] \end{bmatrix}$$

From before we have that **T must be full rank for perfect recoverability** and thus solutions exist only for $N_r \geq N_t$. Moreover size $T = N_r(L_{eq} + 1) \times N_t(L_{eq} + L + 1)$ and thus we have:

- **T is fat:** overdetermined $\rightarrow$ no solution, usually choose so to minimize squared error
- **T is square:** underdetermined $\rightarrow$ one exact solution $w_j^{ZF} = \frac{1}{\sqrt{G}}(T^*)^\dagger d_j$
- **T is tall:** under determined, infinite solutions: choose minimum norm: $w_j^{ZF} = \frac{1}{\sqrt{G}}T(T^*T)^{-1}d_j$

In general recalling [Definition 1 \(Moore Penrose Pseudo Inverse\)](#) unifies all these cases.

$$W^{ZF} = \frac{1}{\sqrt{G}}(T^\dagger)D^*$$

- $N_t > N_r \rightarrow$ Fat $\rightarrow$ no perfect solution $\rightarrow$ Least Squares
- $N_t = N_r \rightarrow$ Square $\rightarrow L = 0$ frequency flat channel $\rightarrow$ a single tap $L_{eq} = 0$ sufficient for perfect ZF
- $N_r > N_t \rightarrow$ Tall $\rightarrow$ many perfect solutions $\rightarrow$ Minimum Norm

If no perfect ZF solution exists, then the error is given by

$$\hat{x}[n] = x[n - \Delta] - \sum_{L=0}^{L_{eq}} W^*[l]y[n - l]$$

where the summation can be rewritten by stacking as usual (y from n to $n - L_{eq}$)

LMMSE Eq

We obtain the following:

$$\bar{W}^{LMMSE} = R_{y[n]}^{-1} R_{y[n]x[n-\Delta]} \quad (2.162)$$

$$= \begin{bmatrix} R_{yy}[0] & R_{yy}[1] & \cdots & R_{yy}[L_{eq}] \\ R_{yy}[1] & R_{yy}[0] & \cdots & R_{yy}[L_{eq}-1] \\ \vdots & \vdots & \ddots & \vdots \\ R_{yy}[L_{eq}] & R_{yy}[L_{eq}-1] & \cdots & R_{yy}[0] \end{bmatrix}^{-1} \begin{bmatrix} R_{yx}[-\Delta] \\ R_{yx}[1-\Delta] \\ \vdots \\ R_{yx}[L_{eq}-\Delta] \end{bmatrix}, \quad (2.163)$$

$$R_{yy}[\ell] = G \sum_{\ell_1=0}^L \sum_{\ell_2=0}^L H[\ell_1] R_{xx}[\ell + \ell_1 - \ell_2] H^*[\ell_2] + N_0 \delta[\ell] I,$$

$$R_{yx}[\ell] = \sqrt{G} \sum_{\ell_1=0}^L H[\ell_1] R_{xx}[\ell + \ell_1].$$

LMMSE Eq

if IID we have

$$W^{MMSE} = \frac{1}{\sqrt{G}} \left(TT^* + \frac{N_t}{SNR} I \right)^{-1} TD$$

$$E = \frac{E_s}{N_t} I - \frac{E_s}{N_t} D^* T^* \left(TT^* + \frac{N_t}{SNR} I \right)^{-1} TD$$

(!possible exam) if there is no noise ($SNR \rightarrow \infty$) then the error goes to 0:

$$E \xrightarrow{SNR \rightarrow \infty} \frac{E_s}{N_t} I - \frac{E_s}{N_t} D^* T^* (TT^* + 0) TD = \frac{E_s}{N_t} I - \frac{E_s}{N_t} \underbrace{D^* T^* \underbrace{(TT^*)}_I}_{I} TD = \frac{E_s}{N_t} I - \frac{E_s}{N_t} I = 0$$

	Equalizer	Error
Linear ZF Scalar	$w^*(z) = \frac{1}{\sqrt{G}} \frac{1}{h(z)}$	
Linear ZF MIMO	$W^{ZF} = \frac{1}{\sqrt{G}} (T^\dagger) D^*$	$\hat{x}[n] = x[n - \Delta] - \sum_{L=0}^{L_{eq}} W^*[l] y[n - l]$
LMMSE (IID and/or frequency flat)	$W^{MMSE} = \frac{1}{\sqrt{G}} \left(TT^* + \frac{N_t}{SNR} I \right)^{-1} TD$	$E = \frac{E_s}{N_t} I - \frac{E_s}{N_t} D^* T^* \left(TT^* + \frac{N_t}{SNR} I \right)^{-1} TD$
Linear ZF Frequency Flat	$W^{ZF} = \frac{1}{\sqrt{G}} H^\dagger$	0

Single-Carrier Frequency Domain Eq

Suppose a channel of L taps and a DFT of length K , $K > L$, then the convolution becomes a circular convolution, that is, x has modulo K inside and h is zero padded.

$$\begin{aligned}
 y[n] &= \sqrt{G} \sum_{l=0}^{K-1} h[l] x[(n-l)_K] \\
 &= \begin{cases} \sqrt{G} \left(\sum_{l=0}^n h[l] x[n-l] + \sum_{l=n+1}^L h[l] x[n+K-l] \right) & n = 0, \dots, L-1 \\ \sqrt{G} \sum_{l=0}^L h[l] x[n-l] & n = L, \dots, K-1 \end{cases}
 \end{aligned}$$

to avoid this mismatch we use a cyclic prefix ($L_c \geq L$):

$$x[n] \neq 0 \forall n \in \{0, \dots, K-1\} \rightarrow \bar{x}[n] = \begin{cases} x[n+K] & n = L_c, \dots, -1 \\ x[n] & n = 0, \dots, K-1 \end{cases} \neq 0, \forall n \in \{0, \dots, L\}$$

so that $x[(n-l)_K] = \bar{x}[n-l], \forall l \in \{0, \dots, L\}$

The output also contains a cyclic prefix but can be discarded as it is redundant information.

Estimation happens by doing the IDFT

$$\hat{\mathbf{x}} = \mathbf{U}^* \text{diag} \left(\frac{1}{\sqrt{G}h[0]}, \dots, \frac{1}{\sqrt{G}h[K-1]} \right) \mathbf{U} \mathbf{y}$$

with $\mathbf{U}$ a Fourier matrix and h the DFT of the channel.

MIMO Case

Vectorize as usual, now the eq is the following:

$$\hat{\mathbf{x}} = \frac{1}{\sqrt{G}} \text{IDFT} \{ H^\dagger[k] \mathbf{y}[k] \}$$

4.4 (!exam candidate) Orthogonal Frequency Division Multiplexing (OFDM)

Base OFDM

OFDM transmits K signals over some subcarriers in parallel. The subcarriers are orthogonal in frequency. Coding takes place in frequency, the IDFT is TXed and L samples in the cyclic prefix are added.

The receiver computes the DFT, removes the cyclic prefix, and can then estimate the signal by ZF.

	Standard	OFDM
ISI	Yes (if frequency selective)	No (if $L \geq L_h$), all subcarriers are frequency flat
Tx Time of block of size T	T	$(K + L)T$, band guard interval LT penalty
Carrier	B	K orthogonal subcarriers of size B/K
Spectral Efficiency	100%	$\frac{K}{K+L}$

Let's do it step by step:

Block of K symbols which is also DFT length, L cyclic prefix, length of channel $L_h \leq L$

Subcarriers $e^{j\frac{2\pi}{K}nk}$ at frequency k/K .

- **Transmitter:**

The block of the signals in frequency is: $x_f[0], \dots, x_f[K-1]$

The IDFT is computed as $x_t[n] = \frac{1}{K} \sum_{k=0}^{K-1} x_f[k] \underbrace{e^{j\frac{2\pi}{K}nk}}_{\text{subcarrier}} \forall n \in \{0, \dots, K-1\}$

The cyclic prefix of length L is added $\bar{x}_t = [x_t(-L), \dots, x_t(-1), x_t[0], \dots, x_t[K-1]]$ where $x_t(-a) = x_t(K-a)$ or also

$$\bar{x}_t[n] = \begin{cases} x_t[n+K] & n = -L, \dots, -1 \\ x_t[n] & n = 0, \dots, K-1 \end{cases}$$

Notice that the IDFT of $x_f[k]$ can also be extended to $n \in \{-L, \dots, K-1\}$ and results in the cyclic prefix output.

Verify (!exam candidate):

$$\begin{aligned} \bar{x}_t[n \in \{-L, \dots, -1\}] &= \frac{1}{K} \sum_{k=0}^{K-1} x_f[k] e^{j\frac{2\pi}{K}nk} \\ &= \frac{1}{K} \sum_{k=0}^{K-1} x_f[k] e^{j\frac{2\pi}{K}nk} \underbrace{e^{j\frac{2\pi}{K}Kk}}_{=1} \\ &= \frac{1}{K} \sum_{k=0}^{K-1} x_f[k] e^{j\frac{2\pi}{K}(K+n)k} = x_t[K+n] \end{aligned}$$

This means that

$$\bar{x}_t[n] = \frac{1}{K} \sum_{k=0}^{K-1} x_f[k] e^{j\frac{2\pi}{K}nk} = \begin{cases} \frac{1}{K} \sum_{k=0}^{K-1} x_f[k] e^{j\frac{2\pi}{K}nk} & = x_t[n] & n = 0, \dots, K-1 \\ \frac{1}{K} \sum_{k=0}^{K-1} x_f[k] e^{j\frac{2\pi}{K}(K+n)k} & = x_t[n+K] & n = -L, \dots, -1 \end{cases}$$

so for example if we need $\bar{x}_t[-1] = \frac{1}{K} \sum_{k=0}^{K-1} x_f[k] e^{j\frac{2\pi}{K}(K-1)k} = x_t[K-1]$.

- **Channel:**

Since for $n \in \{-L, \dots, -1\}$ we need taps of the previous block, we don't calculate them and only keep values for $n \in \{0, \dots, K-1\}$ and therefore $y_t[n]$ doesn't have the cyclic prefix.

Notice that

$$y_t[-1] = \sqrt{G} \sum_{l=0}^{L-1} h_t[l] \bar{x}_t[n-l] + \sqrt{G} h_t[L] \bar{x}_t[-1-L]$$

in general

$$y_t[n] = \begin{cases} \sqrt{G} \sum_{l=0}^L h_t[l] \bar{x}_t[n-l] & n = 0, \dots, K-1 \\ \sqrt{G} \sum_{l=0}^{L+n} h_t[l] \bar{x}_t[n-l] + \sqrt{G} \sum_{l=L+n+1}^L h_t[l] \bar{x}_t[n-l+K] & n = -L, \dots, -1 \end{cases}$$

so by doing the **cyclic prefix removal** we just keep the first part.

This already shows that the channel is a convolution in time so a product in frequency

- **Receiver:**

First we do **cyclic prefix removal** on the received signal $y[n]$ (in time) so to then do the DFT and have all $\tilde{y}[k]$ of the block in frequency.

$$\tilde{y}_f[k] = \sum_{p=0}^{K-1} y_f[p] e^{-j \frac{2\pi}{K} np}$$

Now the full relation starting from $x_f[n]$ is found (I used n instead of k for simplicity in notation):

$$\begin{aligned} \tilde{y}_f[n] &= \sum_{p=0}^{K-1} y_t[p] e^{-j \frac{2\pi}{K} np} \\ &= \sum_{p=0}^{K-1} \sqrt{G} \sum_{l=0}^L h_t[l] \bar{x}_t[p-l] e^{-j \frac{2\pi}{K} np} \\ &= \sum_{p=0}^{K-1} \sqrt{G} \sum_{l=0}^L h_t[l] \frac{1}{K} \sum_{k=0}^{K-1} x_f[k] e^{j \frac{2\pi}{K} (p-l)k} e^{-j \frac{2\pi}{K} np} \\ &= \frac{\sqrt{G}}{K} \sum_{k=0}^{K-1} \left(\sum_{l=0}^{K-1} h_t[l] e^{-j \frac{2\pi}{K} lk} \right) x_f[k] \sum_{p=0}^{K-1} e^{j \frac{2\pi}{K} p(k-n)} \\ &= \sqrt{G} \sum_{k=0}^{K-1} h_f[k] x_f[k] \delta(k-n) \\ &= \sqrt{G} h_f[n] x_f[n] \end{aligned}$$

Lets go step by step:

- First $y[p]$ was rewritten using the DFT formula.
- Then $\bar{x}[n]$ was rewritten using the IDFT formula.
- Then: (i) the constants were extracted (ii) the exponents were rewritten so that

$$\begin{aligned} \exp \left\{ j \frac{2\pi}{K} (p-l)k \right\} \exp \left\{ -j \frac{2\pi}{K} np \right\} &= \exp \left\{ j \frac{2\pi}{K} (pk - lk - np) \right\} \\ &= \exp \left\{ -j \frac{2\pi}{K} lk \right\} \exp \left\{ j \frac{2\pi}{K} p(k-l) \right\} \end{aligned}$$

and then (iii) the sums were rearranged.

- Then notice that (i) $\sum_{l=0}^{K-1} h_t[l] e^{-j \frac{2\pi}{K} lk} = h_f[k]$ is the DFT and (ii) $\sum_{p=0}^{K-1} e^{j \frac{2\pi}{K} p(k-l)}$ is K times the IDFT of $\delta(k-n)$.
- The final expression is done by simplifying the Kronecker delta and noticing that is is zero for all $k \neq n$.

The suffixes are used to indicate if time or frequency.

The whole OFDM stack goes in these steps:

$$x_f \xrightarrow{\text{IDFT+Cyclic Prefix}} x_t \xrightarrow{\text{Channel+Removed Prefix}} y_t \xrightarrow{\text{DFT}} \tilde{y}_f$$

The last formula can simplify in just $x_f \rightarrow \tilde{y}_f$.

The book offers a slightly different version of the proof, starting from $x_f[k]$ and finding $y_t[n]$:

$$\begin{aligned}
 y_t[n] &= \sqrt{G} \sum_{l=0}^L h_t[l] \bar{x}_t[n-l] \\
 &= \sqrt{G} \sum_{l=0}^L h_t[l] \frac{1}{K} \sum_{k=0}^{K-1} x_f[k] e^{j\frac{2\pi}{K}(n-l)k} \\
 &= \frac{\sqrt{G}}{K} \sum_{k=0}^{K-1} \left(\sum_{l=0}^L h_t[l] e^{j\frac{2\pi}{K}lk} \right) x_f[k] e^{-j\frac{2\pi}{K}nk} \\
 &= \sqrt{G} \cdot \frac{1}{K} \sum_{k=0}^{K-1} h_f[k] x_f[k]
 \end{aligned}$$

which is the IDFT of $\tilde{y}_f[k]$: $y_t[n] = IDFT(\hat{y}_f[k]) = IDFT(\sqrt{G}h_f[k]x_f[k])$. Or equivalently the DFT of $y_t[n]$ is $\tilde{y}_f[k]$: $\tilde{y}_f[k] = DFT(y_t[n]) = DFT\left(\sqrt{G} \cdot \frac{1}{K} \sum_{k=0}^{K-1} h_f[k] x_f[k]\right) = \sqrt{G}h_f[k]x_f[k]$

(!exam candidate)

This result ($y_f[k] = \sqrt{G}h_f[k]x_f[k]$) shows that for each subcarrier there is the received signal $\tilde{y}[k]$ that depends only on the channel and signal of that subcarrier.

Estimation:

With absence of noise a simple eq is used:

$$\hat{x}_f[k] = \frac{\tilde{y}_f[k]}{\sqrt{G}h_f[k]}$$

With noise notice that $v_t[n] \approx v_f[k]$ (TODO Appendix C1.6) with the same ZF eq the variance of the output noise becomes

$$\frac{N_0}{G} \frac{1}{|h_f[k]|^2}$$

MIMO OFDM

MIMO-OFDM gives a frequency flat equivalent model, justifying its extensive treatment

With OFDM, the equalizer complexity becomes independent of the channel length, but with a spectral efficiency penalty

$$\mathbf{y}[k] = \sqrt{G}\mathbf{H}[k]\mathbf{x}[k] + \mathbf{v}[k]$$

4.5) Channel Estimation

Intro

For channel estimation some well known signals, called pilot signals $p[0], \dots, p[N_p - 1]$ where $p[\cdot]$ is a $N_t \times 1$ vector. These signals are sent through a L tap channel $\sqrt{G}H[l]$ and from the received signals a estimation on the channel can be made.

To implicitly do the convolutions, build the $N_r \times N_t(L + 1)$ channel matrix

$$\overline{H} = [H[0], \dots, H[L]]$$

And the $N_t(L + 1) \times N_p - L$ pilot matrix

$$\overline{P} = \begin{bmatrix} p[L] & p[L+1] & \dots & p[N_p - 1] \\ p[L-1] & p[L+2] & \dots & p[N_p - 2] \\ \vdots & \vdots & \vdots & \\ p[0] & p[1] & \dots & p[N_p - L - 1] \end{bmatrix}$$

so to have

$$\bar{Y} = \sqrt{G} \bar{H} \bar{P} + \bar{V}$$

alternatively using stacked vectors and Kroenecker product (TODO)

ML and Least-Squares Estimation

Since everything except the noise is known, we can estimate H by finding the ML estimation, which equals to the least squares.

Notice that $y|h \sim \mathcal{N}_{\mathbb{C}}(\sqrt{G}P_{\otimes}h, N_0)$

Moreover the max estimation of the ML has constants that can be ignored so the problem can be reduced as follows:

$$\text{vec}[\hat{H}](y) = \arg \max_h f_{y|h}(y|h) = \arg \max_h \frac{1}{\det(\pi R)} e^{-\frac{|y-\mu|^2}{R}} \rightarrow \arg \min_h |y - \mu|^2$$

By applying the least squares with a linear estimator estimator we have

$$\text{vec}[\hat{H}](y) = \frac{1}{\sqrt{G}}(P_{\otimes}^* P_{\otimes})^{-1} P_{\otimes}^* y \rightarrow \hat{H} = \frac{1}{\sqrt{G}} Y P_{\otimes}^* (P_{\otimes} P_{\otimes}^*)^{-1}$$

where [moore penrose](#) again choses the solution.

- overdetermined: Tall, minimum distance (least squares)
- underdetermined: fat, minimum norm

And the MSE matrix is

$$E = \frac{N_0}{G} (P_{\otimes}^* P_{\otimes})^{-1}$$

TODO simplify LSE

LSMMSE Estimator

Just these formulas and the exercises

$$W_{MMSE} = \sqrt{G} P_{\otimes} (N_0 R_{\text{vec}\hat{H}}^{-1} + G P_{\otimes}^* P_{\otimes})^{-1}$$

$$\text{vec}\hat{H} = W_{MMSE}^* y$$

and the MMSE

$$E = R_{\text{vec}\hat{H}} - G R_{\text{vec}\hat{H}}^* P_{\otimes}^* (G P_{\otimes} R_{\text{vec}\hat{H}} P_{\otimes}^* + N_0 I)^{-1} P_{\otimes} R_{\text{vec}\hat{H}}$$

	ML/Least Squares	LSMMSE
Estimator	$\text{vec}[\hat{H}](y) = \frac{1}{\sqrt{G}}(P_{\otimes}^* P_{\otimes})^{-1} P_{\otimes}^* y$ $\hat{H} = \frac{1}{\sqrt{G}} Y P_{\otimes}^* (P_{\otimes} P_{\otimes}^*)^{-1}$	$\text{vec}\hat{H} = \left((\sqrt{G} P_{\otimes} (N_0 R_{\text{vec}\hat{H}}^{-1} + G P_{\otimes}^* P_{\otimes})^{-1})^* y \right)$
MMSE	$E = \frac{N_0}{G} (P_{\otimes}^* P_{\otimes})^{-1}$	$E = R_{\text{vec}\hat{H}} - G R_{\text{vec}\hat{H}}^* P_{\otimes}^* (G P_{\otimes} R_{\text{vec}\hat{H}} P_{\otimes}^* + N_0 I)^{-1} P_{\otimes} R_{\text{vec}\hat{H}}$

4.6 Exercises (Textbook & Professor)

Example 2.1, 2.2 (Tapped Delay Line).

Consider a signal that propagates from transmitter to receiver incurring a delay τ_0 and experiencing a complex amplitude gain A_0 . Determine c_p, c_b .

We can find the passband version:

$$c_p(\tau) = A_0\delta(\tau - \tau_0)$$

Now the baseband version:

$$\begin{aligned} c_b(\tau) &= g_{B/2}(\tau) * (c_p(\tau)e^{-j2\pi f_c\tau}) \\ &= A_0 B \text{sinc}(B(\tau - \tau_0))e^{-j2\pi f_c\tau_0} \end{aligned}$$

Now consider another ray with a different delay τ_1 and amplitude A_1 :

$$c_p(\tau) = \sum_{i=0}^1 A_i \delta(\tau - \tau_i)$$

and therefore c_b is:

$$c_b(\tau) = \sum_{i=0}^1 A_i B \text{sinc}(B(\tau - \tau_i))e^{-j2\pi f_c\tau_i}$$

Example 2.3, 2.4 (Time Discretized Tapped Delay Line).

Determine the discrete time complex baseband response of 2.1:

Set $T = 1/B$ and the sampling lT , then

$$\begin{aligned} c[l] &= T A B \text{sinc}(B(\tau - \tau_0))e^{-j2\pi f_c\tau_0} \big|_{\tau=lT} \\ &= A \text{sinc}(l - \frac{\tau_0}{T})e^{-j2\pi f_c\tau_0} \end{aligned}$$

by setting the sampling at $lT + \tau_0$ we have

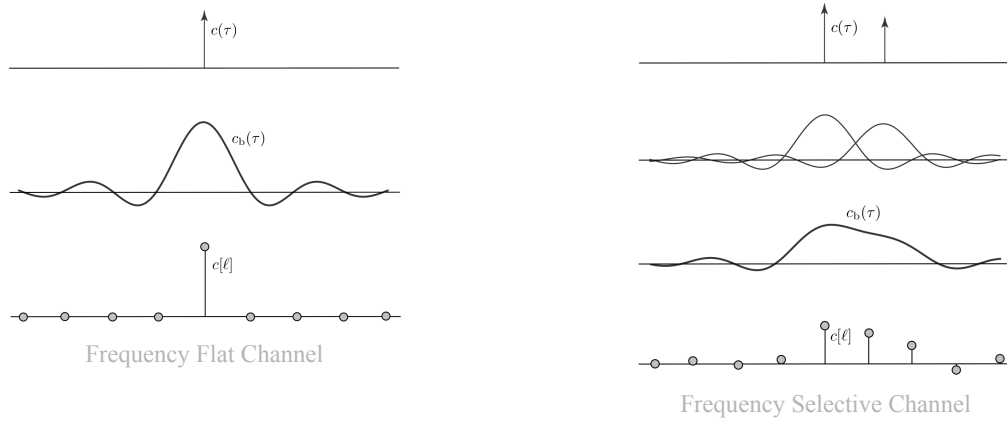
$$c[l] = A \text{sinc}(l)e^{-j2\pi f_c\tau_0} = A\delta(l)e^{-j2\pi f_c\tau_0}$$

In the tapped delay line we have:

$$\begin{aligned} c[l] &= \sum_{i=0}^{N-1} A_i \text{sinc}(B(lT + \phi - \tau_i))e^{-j2\pi f_c\tau_i} \\ &\stackrel{\phi=\tau_0}{=} A_0\delta(l)e^{-j2\pi f_c\tau_0} + \sum_{i=1}^{N-1} A_i \text{sinc}(l - \frac{\tau_i - \tau_0}{T})e^{-j\pi f_c\tau_i} \end{aligned}$$

Which inevitably introduces ISI if at least one τ_i is not a integer multiple of τ_0 .

Recall the properties of the sinc function: $\text{sinc}(a) = a, \forall a \in \mathbb{N} \setminus \{0\}$



In fact the first channel can be obtained from $c(\tau)$ by recalling that $\delta[Tl] = \frac{1}{T}\delta[l]$. While the second channel cannot be correctly obtained as it is not single tap:

$$c(\tau) \xrightarrow{lT+\tau_0} c[l] = A_0\delta(l)e^{-j2\pi f_c\tau_0} \sum_{i=1}^{N-1} A_i\delta(l - \frac{\tau_i - \tau_0}{T})e^{-j2\pi f_c\tau_i}$$

Moreover, to follow the prof exercise 2.8 we have that if $\phi = \tau_1$, then a different channel arises:

$$c[l] = A_1\delta(l)e^{-j2\pi f_c\tau_1} + \sum_i A_i\text{sinc}(l - \frac{\tau_i - \tau_1}{T})e^{-j2\pi f_c\tau_i}, \quad i \in \{1, 2, \dots, N-1\} \setminus \{1\}$$

Example 2.10 (Showing Of precoding Workings).

Let $N_t = N_s = 2$ and suppose that the two signals in $s[n]$ are QPSK distributed. Examine $x[n]$ with a time invariant precoder:

Recall:

$$x[n] = \sqrt{\frac{E_s}{N_t}} F s = \sqrt{\frac{E_s}{N_t}} U \Sigma V^* s$$

First calculate $V^* s = \begin{bmatrix} V_{00}^* s_0 + V_{10}^* s_1 \\ V_{01}^* s_0 + V_{11}^* s_1 \end{bmatrix} = [a, b]^T$. This is the mixing signal matrix, in fact it creates two new signals a, b that are a mix of the original signals.

Then $\Sigma V^* s = [P_0 a, P_1 b]^T$. This is the power allocation matrix (in this case $\Sigma = P^{1/2}$) and allocates different powers to the different signals.

Finally $U \Sigma V^* s = \begin{bmatrix} U_{00} a + U_{01} b \\ U_{10} a + U_{11} b \end{bmatrix} = [\alpha, \beta]^T$. This is the beam matrix and in fact it mixes the two new signals with allocated powers to create the final output signal for each antenna (row).

Then we have:

$$x[n] = \sqrt{\frac{E_s}{N_t}} \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \sqrt{\frac{E_s}{N_t}} \begin{bmatrix} U_{00} P_0 (V_{00}^* s_0 + V_{10}^* s_1) + U_{01} P_1 (V_{01}^* s_0 + V_{11}^* s_1) \\ U_{10} P_0 (V_{00}^* s_0 + V_{10}^* s_1) + U_{11} P_1 (V_{01}^* s_0 + V_{11}^* s_1) \end{bmatrix}$$

This normalizes the average power.

Example 2.15, 2.16 (Two Tap Channel ZF Equalizer).

For a two tap channel where $\sqrt{G}h[0] = 1$, $\sqrt{G}h[1] = -a$ find an IIR equalizer with $\Delta = 1$.

First find transform of $h[n] = \delta(n) - a\delta(n-1) \xrightarrow{z} 1 - az^{-1}$
therefore we have:

$$w(z) = \frac{1}{\sqrt{G}} \frac{z^{-1}}{1 - az^{-1}} \xrightarrow{z^{-1}} w[l] = \frac{1}{\sqrt{G}} a^{l-1} u[l-1]$$

Where an appropriate FIR equalizer can be found by truncating.

Example 2.18, 2.19 (Linear ZF Equalizer For Frequency Flat Channels).

Compute the Zf equalizer for frequency flat channels with $N_r \geq N_t$ (perfect recoverability, tall) and prove the error is null.

Frequency flat ($L=0$), then $T = H$ and $L_{eq} = 0$ and no delay $D = I$, then

$$W^{ZF*} = \frac{1}{\sqrt{G}} H^\dagger$$

The error is found by recalling that H is tall (underdetermined), by moore penrose $H^\dagger = (H^* H)^{-1} H^*$ and thus

$$\sqrt{G} \frac{1}{\sqrt{G}} W^{ZF*} H = (H^* H)^{-1} H^* H = I$$

Example 2.22, 2.23 (Simple OFDM Exercises).

Let $T_{OFDM} = 3.2\mu s$ with $K = 256 = 2^8$ subcarriers and a cyclic prefix of length $L = 64 = 2^6$. Find the chip period T , the passband bandwidth, the subcarrier spacing, and the guard interval of the cyclic prefix.

Recall $T_{OFDM} = (K + L)T \rightarrow T = 10ns \rightarrow B = 1/T = 100MHz$

The spacing is $1/KT = 390.6KHz$ and the guard interval is $LT = 0.64\mu s$. **Impulse response of this duration or less can be handled.**

Now let the subcarrier spacing be $15kHz$ and every $5 MHz$ of bandwidth accomodates $K = 300$ subcarriers. What is the spectral efficiency penalty?

$$P_{min} = \frac{B_{final}}{B_{tot}} = \frac{K \cdot 15 \cdot 10^3}{5 \cdot 10^6} = 0.9$$

The real penalty also includes the guard interval so

$$P = P_{min} \frac{K}{K + L}$$

Example 2.24 (Least Squares Estimatiore For a Frequency Flat SISO Channel).

For a SISO system with frequency flat ($L = 0$) channel estimate the channel and the MMSE with a single pilot $p = e^{j\phi}$.

The single pilot, SISO and frequency flat case transforms the problem into a simple 1D case. In fact $P = p$ and $H = h$ from here also $P_{\otimes} = p$

Recall the solution of the least squares problem:

$$\text{vec}[\hat{H}] = \frac{1}{\sqrt{G}}(P_{\otimes}^* P_{\otimes})^{-1} P_{\otimes}^* y$$

First calculate $P_{\otimes}^* P_{\otimes} = E_s$, then the full matrix multiplications become $\frac{\sqrt{E_s}}{E_s} e^{-j\phi} = \frac{e^{-j\phi}}{\sqrt{E_s}}$.

Then

$$\hat{h} = \frac{e^{-j\phi}}{\sqrt{GE_s}} y$$

Same for the MMSE:

$$E = \frac{N_0}{G} (P_{\otimes}^* P_{\otimes})^{-1} \rightarrow MMSE = \frac{N_0}{GE_s} = \frac{1}{SNR}$$

Also in the case $N_p > 1$ the MMSE becomes

$$MMSE = \frac{1}{N_p SNR}$$

Example 2.29 (LMMSE Estimator For a Frequency Flat SISO Channel).

As before $L = 0$, SISO and with one pilot, then recall

$$W_{MMSE} = \sqrt{G} P_{\otimes} (N_0 R_{\text{vec}\hat{H}}^{-1} + G P_{\otimes}^* P_{\otimes})^{-1}$$

Which in the 1d case becomes:

$$W_{MMSE} = \frac{\sqrt{GE_s}}{N_0 + GE_s} e^{j\phi} = \frac{\sqrt{GE_s}/N_0}{1 + GE_s/N_0} \frac{\sqrt{GE_s}}{\sqrt{GE_s}} e^{j\phi} = \frac{SNR}{1 + SNR} \frac{e^{j\phi}}{\sqrt{GE_s}}$$

then $\text{vec}\hat{H} = W_{MMSE}^* y$ means that

$$\hat{h} = \frac{SNR}{1 + SNR} \frac{e^{j\phi}}{\sqrt{GE_s}} y$$

The MMSE then becomes, $E = R_{\text{vec}\hat{H}} - G R_{\text{vec}\hat{H}}^* P_{\otimes}^* (G P_{\otimes} R_{\text{vec}\hat{H}} P_{\otimes}^* + N_0 I)^{-1} P_{\otimes} R_{\text{vec}\hat{H}}$ and in 1d (works for $N_p \geq 1$)

$$MMSE = 1 - GE_s (GE_s + N_0)^{-1} = \frac{1}{1 + N_p SNR}$$

Which is *always smaller than the least squares case*

LMMSE ESTIMATION COINCIDES WITH LEAST SQUARES IN IID MIMO

if $P_{\otimes}$ Is unitary, then $P_{\otimes} P_{\otimes}^* = \frac{E_s}{N_t} I$ and thus

$$\begin{aligned}
E &= I - GP_{\otimes}^*(GP_{\otimes}P_{\otimes}^*)^{-1}P_{\otimes} = I - GP_{\otimes}^*\left(\frac{GE_S}{N_t} + N_0\right)^{-1}IP_{\otimes} \\
&= I - GP_{\otimes}^*P_{\otimes}\left(\frac{GE_S}{N_t}\left(1 + \frac{N_t}{SNR}\right)\right)^{-1} \\
&= I - \frac{GE_S}{N_t}\frac{1}{\frac{GE_S}{N_t}\left(1 + \frac{N_t}{SNR}\right)}I \\
&= \frac{1}{1 + \frac{SNR}{N_t}}I
\end{aligned}$$

Exercise 2.9 (Different Channel Representations).

Let $N_t = 1$ and $N_r = 2$ with

$$\begin{aligned}
c^{(0,0)}(\tau) &= A_0\delta(\tau - \tau_0) \\
c^{(1,0)}(\tau) &= A_1\delta(\tau - \tau_1) + A_2\delta(\tau - \tau_2)
\end{aligned}$$

- Determine the pseudo baseband channel:

$$\begin{aligned}
c_{pb}^{(0,0)}(\tau) &= A_0\delta(\tau - \tau_0)e^{-j2\pi f_c\tau_0} \\
c_{pb}^{(1,0)}(\tau) &= A_1\delta(\tau - \tau_1)e^{-j2\pi f_c\tau_1} + A_2\delta(\tau - \tau_2)e^{-j2\pi f_c\tau_2}
\end{aligned}$$

- Determine the baseband channel:

$$\begin{aligned}
c_b^{(0,0)}(\tau) &= BA_0\text{sinc}(B(\tau - \tau_0))e^{-j2\pi f_c\tau_0} \\
c_b^{(1,0)}(\tau) &= BA_1\text{sinc}(B(\tau - \tau_1))e^{-j2\pi f_c\tau_1} + BA_2\text{sinc}(B(\tau - \tau_2))e^{-j2\pi f_c\tau_2}
\end{aligned}$$

- Determine the complex baseband channel in frequency domain:

$$\begin{aligned}
c_b^{(0,0)}(\tau) &= A_0\text{rect}\left(\frac{\tau}{B}\right)e^{-j2\pi(f_c+f)\tau_0} \\
c_b^{(1,0)}(\tau) &= A_1\text{sinc}\left(\frac{\tau}{B}\right)e^{-j2\pi(f_c+f)\tau_1} + A_2\text{sinc}\left(\frac{\tau}{B}\right)e^{-j2\pi(f_c+f)\tau_2}
\end{aligned}$$

By recalling that $\text{sinc}(Bx) \xrightarrow{\mathcal{F}} \frac{1}{B}\text{sinc}\left(\frac{f}{B}\right)$ and $x(t - t_0) \xrightarrow{\mathcal{F}} x(f)e^{-j2\pi ft_0}$

- Determine time discrete baseband channel:

$$\begin{aligned}
c_b^{(0,0)}(l) &= A_0\text{sinc}(l - B\tau_0)e^{-j2\pi f_c\tau_0} \\
c_b^{(1,0)}(l) &= A_1\text{sinc}(l - B\tau_1)e^{-j2\pi f_c\tau_1} + A_2\text{sinc}(l - B\tau_2)e^{-j2\pi f_c\tau_2}
\end{aligned}$$

✓ **Todo**

all exercises on ZF

5) Channel modeling

Basics

Many non-LoS paths exist (diffraction, reflection, refraction, scattering).

Two approaches are used to model the channel:

1. Deterministic: develop and use a site-specific description of the environment
2. Stochastic: channel models are random processes inspired by measurements

Large Scale Phenomena

The link budget formula in dB gives useful insights:

$$P_r[dB] = P_t[dB] + \underbrace{G_t[dB] + G_r[dB] - (L_p[dB] + \chi[dB])}_{\text{large scale gain } G}$$

Where the large scale gain G [dB] is the gain used until now, in linear scale it becomes:

$$G = \frac{G_t G_r}{L_p \chi}$$

Moreover χ is a zero mean gaussian rv representing shadowing.

While the path loss is:

$$L_P(D) = K_{ref} \left(\frac{D}{D_{ref}} \right)^\eta \rightarrow L_P(D)[dB] = K_{ref}[dB] + 10\eta \log_{10} \left(\frac{D}{D_{ref}} \right)$$

It is possible that many functions with different parameters are used to model a channel with piecewise linear behavior.

The **path loss in free space** is:

$$L_P(D)[dB] = 20 \log_{10} \left(\frac{4\pi D}{\lambda_c} \right)$$

Small-Scale Fading

Recall the [tapped delay line model](#):

Suppose $x(t)$ is a passband sinusoid, then with a narrowband signal with $\max_q \tau_q \gg 1/B$ the delays act as a phase shift:

$$x(t) = e^{j2\pi f_c t} \xrightarrow{\text{channel}} y(t) = \sum_q A_q e^{j2\pi f_c (t - \tau_q)} = \sum_q A_q e^{-j2\pi f_c \tau_q} x(t)$$

clearly the channel can be simplified as

$$c(\tau) = \sum_q A_q e^{-j2\pi f_c \tau_q} \delta(\tau)$$

which can be rewritten as a **large scale component** $\sqrt{G}$ and a **small scale component** h then $c = \sqrt{G}h$.

Model the small-scale channel response as a locally stationary complex rv $h = |h|e^{j\angle h} \sim \mathcal{N}_{\mathbb{C}}(0, 1)$, then (TODO Appendix C 1.9) the channel has these 3 properties:

- Uniform Phase: $\phi = \angle h \sim U(0, 2\pi)$
- Rayleigh Magnitude: $f_{|h|}(\xi) = \xi e^{-\frac{1}{2}\xi^2}$
- Exponential magnitude squared: $f_{|h|^2}(\xi) = e^{-\xi}$

If the multi path components dominate over the LoS, then rice fading can be used:

$$f_{|h|}(\xi) = 2(K+1)\xi e^{-(K+1)\xi^2-K} I_0 \left(2\sqrt{K(K+1)}\xi \right)$$

where

- $K = 0$: rice equals Rayleigh
- $K \rightarrow \infty$: rice is deterministic

Power Angle Spectrum

The PAS works with the **azimuth** angle and describes the Rx power as a function of the angle.

By integrating the PAS in the angle the total received power is obtained.

Example 3.4 (Clarke-Jake Uniform PAS).

The uniform PAS is

$$\mathcal{P}(\theta) = \frac{1}{2\pi}, \theta \in [-\pi, \pi)$$

and therefore the total RX power is

$$P = \int_{-\pi}^{\pi} \mathcal{P}(\theta) d\theta = 1$$

Example 3.7 (Truncated Laplacian PAS).

The truncated Laplacian PAS is

$$\mathcal{P}(\theta) = \frac{K_L}{\sqrt{2}\sigma} \exp \left\{ - \left| \frac{\sqrt{2}(\theta - \mu)}{\sigma} \right| \right\}, \theta \in [\mu - \pi, \mu + \pi)$$

Where

$$K_L = \frac{1}{1 - \exp \left\{ -\sqrt{2}\pi/\sigma \right\}}$$

And in fact the integral, after many manipulations, is normalized to 1

$$P = K_L \left(1 - \exp \left\{ -\sqrt{2}\pi/\sigma \right\} \right) = 1$$

Space Correlation

Again, supposing x a complex exponential being received at a given location

$$y(t) = \sqrt{G} \int_{2\pi} \sqrt{\mathcal{P}(\theta) G_r(\theta)} x(t - D(\theta)/C) d\theta$$

$$y = \sqrt{G} \int_{2\pi} \sqrt{\mathcal{P}(\theta) G_r(\theta)} x e^{-j\phi(\theta)} d\theta$$

Where $\phi(\theta) = 2\pi D(\theta)/\lambda_c$

At a second location distanced Δ_d from the first one we have that $\phi'(\theta) = \phi(\theta) + 2\pi\Delta_d \cos(\theta)/\lambda_c$

Their correlation in the case of **phase uniformly distributed in $-\pi, \pi$ is**

$$R_h(\Delta_d) = \mathbb{E}[h_0 h_1^*] = \int_{-\pi}^{\pi} \mathcal{P}(\theta) G_r(\theta) e^{j2\pi\Delta_d \cos(\theta)\lambda_c} d\theta$$

This should be put to 0 for optimal reception.

For the Clarke-Jake PAS this equals to

$$D_C = 0.38\lambda_c \approx 0.5\lambda_c$$

Also

$$T_c \approx D_c/v$$

Time Correlation

Let the **maximum doppler shift** be

$$\nu_M = \frac{v}{c} f_c$$

and the **doppler shift**

$$\nu = \nu_M \cos(\theta)$$

The previous result can be adopted as a signal sent while moving, so by integrating over the max doppler shift we have

$$R_h(\Delta_t = \Delta_d/v) \xrightarrow{\mathcal{F}} S_h(\nu) = \frac{\mathcal{P}(-\arccos(\nu/\nu_M)) + \mathcal{P}(\arccos(\nu/\nu_M))}{\sqrt{\nu_M^2 - \nu^2}}, \nu \in [-\nu_m, \nu_m]$$

And for the Clarke Jake PAS this equals to the **horned doppler spectrum**

$$R_h(\Delta_t) = J_0(2\pi\nu_M\Delta_t) \xrightarrow{\mathcal{F}} S_h(\nu) = \frac{1}{\pi\sqrt{\nu_M^2 - \nu^2}}, \nu \in [-\nu_m, \nu_m]$$

	General	Clarke-Jakes
Space Selectivity: PAS	$\mathcal{P}(\theta)$	$\mathcal{P}(\theta) = \frac{1}{2\pi}$
Space Correlation: phase $\phi(\theta) \sim U[-\pi, \pi]$	$R_h(\Delta_d) = \mathbb{E}[h_0 h_1^*] = \int_{-\pi}^{\pi} \mathcal{P} G e^{j2\pi\Delta_d \cos(\theta)\lambda_c} d\theta$	$R_h(\Delta_d) = J_0(2\pi\Delta_d/\lambda_c)$ $D_c = 0.38\lambda_c$
Time Correlation: Time and Frequency	$R_h(\Delta_t = \Delta_d/v)$ $S_h(\nu) = \frac{\mathcal{P}(-\arccos(\nu/\nu_M)) + \mathcal{P}(\arccos(\nu/\nu_M))}{\sqrt{\nu_M^2 - \nu^2}}, \nu \in [-\nu_m, \nu_m]$	$R_h(\Delta_t) = J_0(2\pi\nu_M\Delta_t)$ $S_h(\nu) = \frac{1}{\pi\sqrt{\nu_M^2 - \nu^2}}$

Coherence Time

The coherence time is the measure of the time duration over which the wireless channel is considered "static" or constant.

$$T_c \gg T_s \quad T_c = \frac{1}{2\nu_M} \propto \frac{1}{\nu_M}, \quad \nu_m \ll B$$

Discrete Time Model

In this case

$$R_h(lT) = \frac{1}{T} S_h\left(\frac{\nu}{T}\right)$$

In the block fading model the number of single carrier symbols per block is

$$N_c = \frac{T_c}{T} = BT_c \xrightarrow{\text{IID Blocks}} \frac{1}{2\nu_m T}$$

Moreover in OFDM we have that

$$N_c = \frac{K}{K+L} T_c B_c \approx T_c B_c$$

Frequency selective Channels, Power Delay Spread

If the channel has large enough bandwidth then it is not possible to model a block in which the channel is constant. In this case the channel is called frequency selective and it is not multiplicative anymore, but convolutional:

Definition 6 (Delay Spread).

The maximum delay between the multi path signal is called delay spread:

$$\max_{i,j} |\tau_i(t) - \tau_j(t)| = \tau_m \triangleq \text{Delay Spread}$$

If the Delay spread is much smaller than the symbol time, the channel is frequency flat

$$\tau_m \ll B_u^{-1} \approx T_s \implies \tau_i(t) \approx \tau_j(t) \approx \hat{\tau} \quad \forall i, j \rightarrow h(\tau) = h\delta(\tau)$$

Here **gain is constant across frequency and the estimation of $g(t)$ suffices to compensate for phase and amplitude displacements**

However if the delay spread is not much smaller than the symbol time, the channel receives signals from different blocks.

The **received power as a function of delay (normalized)** is called **Power Delay profile (PDP)** and can be interpreted as a PDF.

$$S_h(\tau) = \mathbb{E}[|h(\tau)|^2]$$

It has the following mean and variance:

$$\begin{aligned} \mu_\tau &= \int \tau S_h(\tau) d\tau & \longrightarrow \sum_{q=0}^{Q-1} \tau_q P_q \\ \sigma_\tau^2 &= T_d = \sqrt{\int (\tau - \mu_\tau)^2 S_h(\tau) d\tau} & \longrightarrow \sqrt{\sum_{q=0}^{Q-1} (\tau_q - \mu_q)^2 P_q} \end{aligned}$$

and then the frequency flat condition can be rewritten as

$$B_c \gg B \text{ or } T_d \ll T$$

Recap of all equivalent frequency flat conditions:

$$\boxed{\text{Frequency Flat iff: } \tau_m \ll T_s \iff T_d \ll T_s \iff B_c \gg B}$$

If the symbols are uncorrelated, then the fourier transform of the PDP is the frequency domain correlation $R_h(\Delta_f)$ (depends only on Δ_f , wide sense stationary)

$$R_h(\Delta_f) = \int S_h(\tau) e^{j2\pi\Delta_f\tau} d\tau$$

if $\mathbb{E}[h(\tau_0)h(\tau_1)] = 0$ uncorrelated scattering
And from here we have that

$$B_c \propto \frac{1}{T_d}$$

Moreover wireless fading is **underspread**:

$$T_d \ll T_c, \nu_M \ll B_c \rightarrow B_c T_c \gg 1$$

5.2) Exercises (Textbook & Professor)#

Example 3.13 (Clarke-Jakes PAS Coherence Time and Distance).

Let $f_c = 2 \text{ GHz}$ and a velocity $v = 5 \text{ km/h}$, we have:

$$\lambda_c = \frac{c}{f_c} = 15 \text{ cm} \rightarrow D_c = 5.7 \text{ cm}$$

And the coherence time is (recall $\text{m/s} = 3.6 \cdot \text{km/h}$)

$$T_c = \frac{1}{2\nu_m} = \frac{1}{2\frac{v}{c}f_c} = 41 \text{ ms} = \frac{D_c}{0.76 \cdot v} \approx \frac{D_c}{v}$$

Example 3.23 (Continuous PDP distribution Calculation).

Let the PDP be

$$S_h(\tau) = K_e \exp\{-4 \cdot 10^5 \tau\}, \tau \in [0, 10] \mu s$$

First set K_e so to normalize the PDP:

$$1 = \int_0^{10 \cdot 10^{-6}} S_h(\tau) d\tau \approx 2.45 \cdot 10^{-6} K_e \rightarrow K_e = 4.08 \cdot 10^5$$

The mean is:

$$\begin{aligned} \mu_\tau &= \int_0^{10} \tau K_e \exp\{-4 \cdot 10^5 \tau\} d\tau = K_e \left[-\frac{e^{-4 \cdot 10^5 \tau}}{4 \cdot 10^5} \left(\tau + \frac{1}{4 \cdot 10^5} \right) \right] \Big|_0^{10} \\ &\approx 5.68 \cdot 10^{-12} K_e = 2.31 \mu s \end{aligned}$$

And the RMS delay spread can be calculated in closed form but just the result is shown:

$$T_d \approx 2.08 \mu s$$

This means that a symbol time $T \gg 2.08 \mu s$ is required for a frequency flat approximation.

$$B_c \approx \frac{1}{T_d} = 480 \text{ kHz}$$

and therefore supposing we want at least 1000 symbols as the underspread at 2 GHz, it is possible to go as fast as

$$B_c T_c = \frac{B_c}{2\nu_M} = 1000 \rightarrow \nu_m = \frac{f_c v}{c} = \frac{B_c}{2000} \rightarrow v = \frac{c B_c}{2000 f_c} = 36 \text{ m/s} = 130 \text{ km/h}$$